

active

Active Strategies

Development approaches that have proven effective.

- Explore existing codebase patterns before writing new code
 - Write tests alongside implementation
 - Keep changes small and reviewable
-

Build & CI — six named gates, run identically local and remote

README.md:56-71 and .github/workflows/ci.yml are kept in exact correspondence (“The commands below mirror .github/workflows/ci.yml exactly”). Run from the repository root:

```
cd _tools/gen && go test ./... && cd -          # 1 Go unit tests
      (generator)
bash _tools/audit-hardcoding.sh
      # 2 hardcoding audit (builds gen)
bash _tools/translate/reproducibility-selftest.sh # 3 translate
      reproducibility
bash _tools/portfolio/self-validate.sh           # 4 portfolio $1.1
      data integrity
bash _tests/run-harness-selfvalidation.sh
      # 5 visual $11.4.170 + export $11.4.168
# 6 Playwright chromium, excluding the LIVE all-language crawl:
cd _tests && npm ci && npx playwright install chromium
npx playwright test --project=chromium --grep-invert "all-
      language"
```

Toolchains (CI-matched): **Go 1.26, Node 20, Ruby 3.3 + Bundler, poppler-utils, tesseract-ocr** (CLAUDE.md:132-134, README.md:45-52). milosvasic.ru/_site must be built with Jekyll before gate 6 — it is git-ignored in that submodule. vasic.digital needs no build.

The all-language crawl is deliberately excluded from per-run CI — it is designed to run LIVE against production, from the deploy path, not per commit (README.md:76-78).

Testing — self-validation over assertion

The harness proves itself before it judges anything: each self-validate drives a golden-good fixture (expect PASS) and a golden-bad fixture (expect FAIL) and exits 0 only when both hold (`_tests/GATES.md` traceability table). Gate literals (CM-HOST-RENDERED-UI-VISUAL-PROOF, CM-EXPORTED-DOC-VISUALLY-VALIDATED, CM-OPENDESIGN-UI-SYSTEM) are embedded as GATE: echo lines inside the satisfying script so a reviewer can grep from doc → runnable check → evidence.

`_tests/TEST-TYPES.md` maps all 14 §11.4.169 test types to PRESENT / ADDED / N/A-with-reason. Stack: @playwright/test, @axe-core/playwright, pixelmatch + pngjs (`_tests/package.json`).

CM-OPENDESIGN-UI-SYSTEM is **not green** — no self-validated analyzer and no “zero non-token literal” lint exists yet (`_tests/GATES.md`, honest-partial section). Do not report it as covered.

Deploy — incremental, idempotent, PASS-gated

`bash _tools/deploy-langs.sh (--dry-run / -n previews):`

1. A language counts COMPLETE only when its PASS verdict count under `_tests/evidence/translate-new/<lang>/` reaches the source-doc count.
2. Builds gen once with the © year pinned via `-ldflags`.
3. Regenerates EN, then every COMPLETE language, into **both** site submodules.
4. Rebuilds PDFs (skips with a warning if pandoc/weasyprint are missing).
5. Rebuilds the Jekyll `_site`.
6. Commits and pushes each site **only if something changed**.
7. Validates the LIVE sites with the exhaustive all-language link/sitemap crawl.

Idempotent by construction: deterministic output means an unchanged run is a git no-op. Safe to run repeatedly while a translation batch is still writing.

Governance — pointer inheritance, verified

Five root carriers (CLAUDE.md, AGENTS.md, QWEN.md, GEMINI.md, Constitution.md) each open with `## INHERITED FROM`. The four agent carriers are byte-identical from line 24 down — verify with `tail -n +24 <file> | sha256sum` (Constitution.md header table). Pointer form is used instead of `@import` because the corpus is 654 KB + 1.53 MB, ~165k tokens per session (CLAUDE.md:54-66).

Verifiers: `tests/test_constitution_inheritance.sh` and `scripts/verify-all-constitution-rules.sh` (last recorded sweep: **58 gates — 37 PASS / 21 FAIL / 0 ERROR**, exit 1, all 21 failures classified as known-excluded or upstream in Constitution.md).

Commits — the wrapper, not raw git

`commit "<message>"` resolves on PATH to `<toolkit>/Upstreamable/commit`, which chains `git add . + git commit` then `push_all.sh`; `push_all.sh` reads the `upstreams/*.sh` recipes, and this root has a tracked `upstreams/GitHub.sh` exporting `UPSTREAMABLE_REPOSITORY=git@github.com:milos85vasic/vasic.git` (CLAUDE.md:174-181). Because it runs `git add .`, keep `.gitignore` accurate.

Semantic tooling

Lumen (local embeddings via ollama, model `ordis/jina-embeddings-v2-base-code` — `scripts/lumen-reindex.sh:45`) plus CodeGraph (`.codegraph/codegraph.db`). `_tests/evidence/` is excluded from the Lumen index via `.lumenignore`; everything else, submodules included, is indexed. Helpers: `scripts/lumen-reindex.sh`, `scripts/lumen-index-doctor.sh`.